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Enhancing Geological Storage of Hydrogen in Subsurface Caverns

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Abstract

Global hydrogen (H_2) production is projected to reach ~500 Mt/year by 2050. Large-scale storage of H_2 is considered a pre-requisite for the hydrogen economy. Subsurface H_2 storage including in salt caverns is seen as a viable option that supports large-scale storage and on-demand access to large-quantity energy. Here we present a novel methodology to use adsorbents in salt caverns for maximizing the amount of working gas that can be stored under a given geological scenario, with constrained pressure/temperature and H_2 phase behavior. We evaluated the adsorption and desorption capabilities of various sorbent materials under different P-T conditions to determine most optimal sorbents for hydrogen storage. Scaled caverns filled with these sorbents will be subjected to representative P-T conditions in shallow caverns to investigate the hydrogen storage enhancement and also to quantify the potential impact of sorbents on cavern operations. Our results indicate that the microporous activated carbon material could be a suitable candidate for underground hydrogen storage. It demonstrates that under shallow cavern conditions, geological hydrogen storage capability can be potentially enhanced by up to 15% in sorbent-filled caverns when compared with traditional caverns without filling the sorbents. We evaluated multiple sorbent materials and noted that some of the commercially available porous sorbents could be used for this application with favorable economics over the lifetime of the cavern. We further suggest that our methodology can be potentially applied to the current operational storage sites, with the maximum enhancement for caverns with a shallower depth of < 600 m. Subsequent efforts will focus on optimizing and modifying the sorbents for improved commercial feasibility. Operational requirements place a strong constraint on the volume of working gas in a cavern. Maximizing the amount of working gas in each cavern could aid in reducing the operational footprint and associated costs by reducing the number of caverns needed for storage. Our results suggest that mechanical support from sorbents will also allow access to larger volumes of stored gas compared to caverns without sorbent. The proposed technology is at the intersection of novel chemistry and geological storage operations. It presents a novel way to significantly reduce the number of caverns needed while simultaneously reducing the operating costs over the life cycle of caverns.

Introduction

Net-zero transition in world energy systems aiming at a more sustainable future involves increasing usage of renewable energy sources and low-carbon feedstock (Fankhauser et al., 2022). Hydrogen (H_2), when compared with other energy vectors, has attracted special attention not only because it can be used as high-quality, renewable, carbon-free feedstock, but also as a favorable intermediate medium or carrier for energy transportation and storage to mitigate the intermittent nature of renewable energy generations (e.g., Tarkowski, 2019; Kannah et al., 2021; Williams et al., 2022). In the last decade, hydrogen demand has significantly increased and reached a global demand of approximately 100 million metric tons per year (Mt/year), with a further projection of ~500 Mt/year by 2050 according to the International Energy Agency's (IEA) Net Zero Roadmap. More than 30 countries have published hydrogen energy roadmaps and Saudi Aramco has also announced its ambition to achieve operational net-zero Scope 1 and Scope 2 greenhouse gas emissions across its wholly-owned operated assets by 2050. The rapidly increasing hydrogen economy leads to continuous R&D efforts on developing sustainable, clean, renewable, and economic hydrogen sources such as nuclear, biomass, wind, tidal, solar and geothermal energy. However, renewable energy generation is commonly intermittent and largely affected by daily or seasonally fluctuating natural events (e.g., sunlight level, tidal fluctuation and wind force) and limited by the accessibility to some critical local resources (e.g., radioactive mine, geothermal spot and microbial community) (e.g., Zivar et al., 2021). When further combined with annually varying, but more steady energy demand, renewable energy excesses or deficits can occur, exerting significant burden on the energy supply network. For example, in California, the spring of 2020 produced an energy surplus of up to 300,000 MWh from renewable sources during each month (EIA, 2021), but on August, an energy shortage occurred due to the sudden increase of energy demand (California ISO, 2021). In order to balance the gap between energy supply and consumption, excess energy generated from renewable sources can be converted to hydrogen through water electrolysis by using the excess produced electricity and then stored temporarily for later usage during periods of high energy demand. It is noteworthy that the loss during the energy transportation is suggested to be much lesser if using a gaseous carrier (<0.1%) compared to using the power network such as electricity grid (~8%) (Zivar et al., 2021). Therefore, on-demand access to (local) large quantities of hydrogen will be a critical component to stabilize the energy network and reduce the overall operational cost caused by the regional and/or seasonal fluctuations of both hydrogen supply especially from the renewable sources and demand across different end-use applications.

However, the storage of hydrogen is challenging especially for mid- to long-term, large-scale hydrogen storage because of its low density and viscosity, high diffusivity and high reactivity with metals. Conventionally, hydrogen is stored, generally in small scale, above the surface as compressed gas in high-pressure gas cylinder/pipeline, liquid hydrogen in cryogenic tank or solid metal hydrides via chemical reactions with certain metals or alloys. Although hydrogen has one of the highest energy density values per unit mass (120 MJ/Kg vs. CH_4 45 MJ/kg; Aftab et al., 2022), being the lightest molecule, it has a very low mass density (H_2 0.082 kg/m³ vs. CH_4 0.657 kg/m³ at 25 °C and 1 atm), indicating that a larger volumetric storage capacity of hydrogen is required to deliver the same energy amount of natural gas such as methane (Lanz et al., 2001; Heinemann et al., 2021a; Navaid et al., 2023; Malki et al., 2024) (Figure 1). Furthermore, hydrogen is highly corrosive to metals particularly through hydrogen embrittlement and diffusion processes, leading to increasing safety concern on hydrogen leakage caused by metal failure especially during mid- to long-term storage in metal tanks (e.g., Hassanpouryouzband et al., 2021). Although solid-state hydrogen storage is considered a safer way due to the lower operational pressure and leakage risk when compared with liquid/gaseous hydrogen storage, it requires a heavy-mass storage system that is commonly associated with limited cycle life, high operational cost and slow hydrogen release kinetics, resecting its overall benefits for wider storage applications. Due to these limitations, hydrogen stored above the surface is generally small in scale and has limited applicability (Zhu et al., 2023) and hence, considered only as a temporary solution

for short-term storage with limited capacity (e.g., Pellow et al., 2015; Mouli-Castillo et al., 2021; Williams et al., 2022).

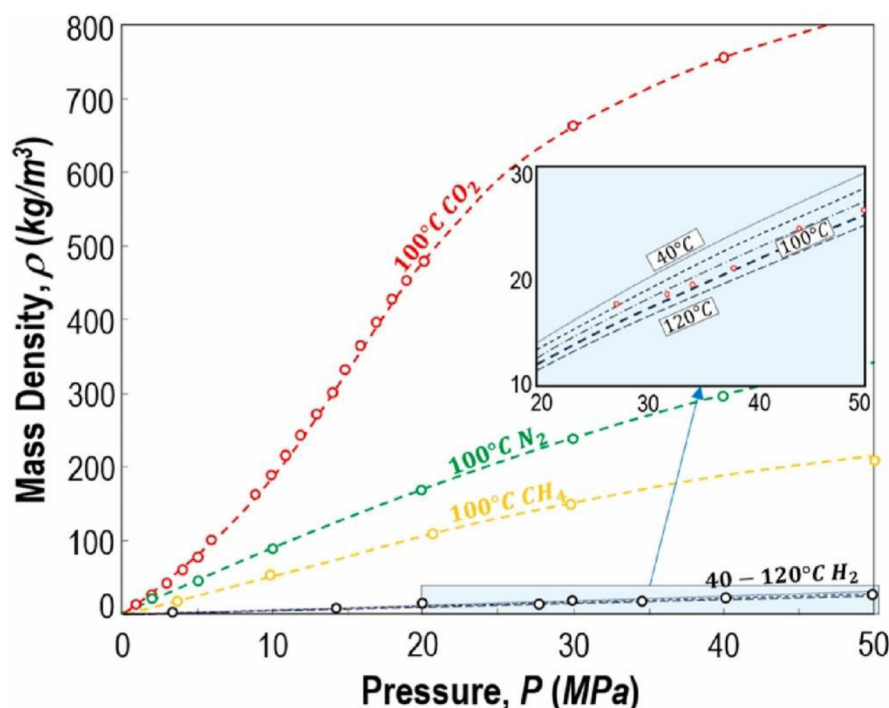


Figure 1—Variation of H₂, CH₄, N₂ and CO₂ density as a function of pressure at different temperatures (from Malki et al., 2024). Note the much lower mass density of hydrogen when compared to other gases at similar P-T conditions.

Underground Hydrogen Storage (UHS) in Geological Formations

One of the major challenges for reliable, on-demand access to hydrogen energy is the availability of low-cost, high-volume storage systems that can support multiple injection-withdrawal cycles in any given year. Underground geological formations support storing such large amounts of hydrogen gas (Zivar et al., 2020). In fact, underground hydrogen storage (UHS) is suggested to be the cheapest way for large-scale gaseous hydrogen storage (Taylor et al., 1986) and the first field test was successfully accomplished over a century ago. Particularly in recent years, UHS has gained significant attention mainly driven by the demand of clean energy sources for net-zero transition goals. Even so, the experience to date with UHS in porous geological formations is still limited to a few storage sites, but the extensive experience for subsurface storage of CO₂, oil and gas, compressed air and fluid waste could be used as analogs or guidance for UHS (e.g., Panfilov, 2016; Miocic et al., 2023; Schultz et al., 2023).

Gaseous hydrogen can be stored in porous geological formations including depleted oil and gas reservoirs (e.g., Lysy et al., 2021; Kanaani et al., 2022; Sekar et al., 2023), saline aquifers (e.g., Heinemann et al., 2021a, b; Raad et al., 2022) and caverns (e.g., Ozarslan, 2012; Caglayan et al., 2020; Navaid et al., 2023) for both short- and long-term storage purposes. Nine UHS sites have been constructed globally, providing a total hydrogen storage capacity of > 600 million cubic meters (see a summary in Zhu et al., 2023). Most hydrogen, however, is stored as gas mixtures in the subsurface with a maximum hydrogen content of up to 62%, only except for several salt caverns where pure hydrogen (>95%) is stored. Despite hydrogen loss could occur at variable levels depending on the chemical and physical properties of the storage systems (Tarkowski, 2019; Zhu et al., 2023; Shahriar et al., 2024), the UHS still provides a local solution for multiple, on-demand accesses to large quantities of hydrogen, reduces exposure to atmospheric oxygen for safer operation, and enables long-term storage viability with high energy storage density (Figure 2).

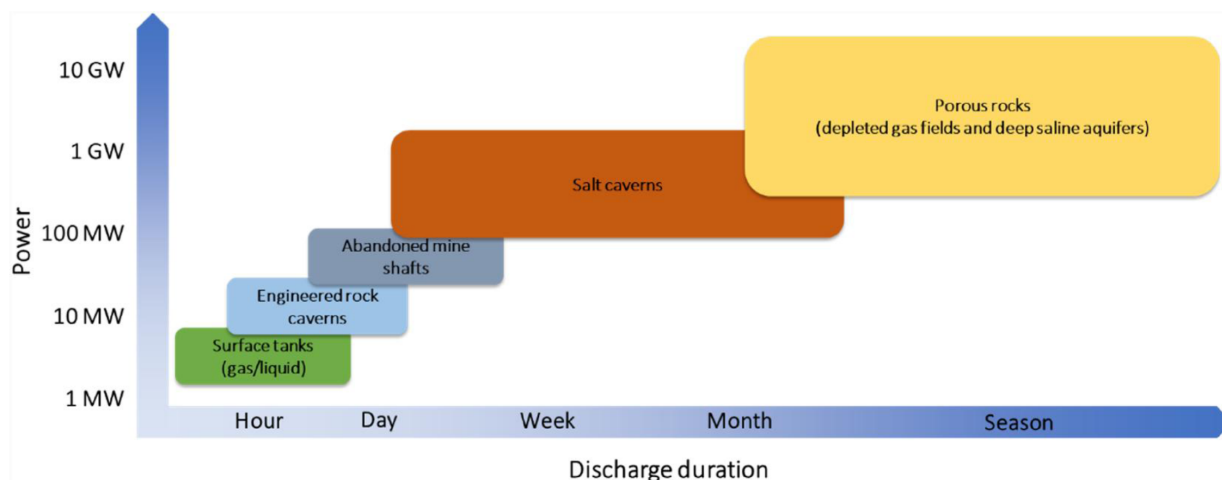


Figure 2—Discharge duration vs. storage power plot for different UHS methods (from [Miočic et al., 2023](#)).

In general, hydrogen is stored in the porous media or void spaces together with cushion gas (e.g., N_2 or CH_4). During gas withdrawal, a certain amount of gases including hydrogen has to be left to maintain the required pressure and deliverability rate in the media ([Oldenburg, 2003](#)). Although the presence of cushion gas helps to minimize the hydrogen loss, extra separation process is needed during or after the withdrawal period to separate hydrogen from the cushion gas. Salt caverns, however, are currently the only commercially proven subsurface storage method implemented at scale to store pure hydrogen ([Williams et al., 2022](#)). When compared with other UHS systems, salt caverns have numerous benefits and is predicted to be the dominant UHS method in the future due to its enhanced self-healing properties, minimum cushion gas requirement, high injection and withdrawal rates and potentials for multiple injection-withdrawal cycles in a relatively short time period ([Ozarslan, 2012](#); [Caglayan et al., 2020](#); [Lankof et al., 2022](#); [Małachowska et al., 2022](#); [Tackie-Otoo and Haq, 2024](#)). As suggested by the International Energy Agency (IEA, 2022), among the 27 UHS sites to be built worldwide, 19 are salt caverns. Due to the very small molecular size, hydrogen diffuses easily in depleted porous gas/oil reservoirs and saline aquifers. The underground storage of hydrogen therefore requires very good sealing property under high hydrogen entry pressures. Rock salt compared with other rock types has very low permeability commonly without pores and has excellent creep properties that can absorb large deformation to minimize or resist crack growth ([Wu et al., 2020](#)). Damaged rock salt can also heal under certain conditions, reducing the hydrogen leakage during mid- to long-term storage. Although smaller than depleted reservoirs or aquifers in terms of potential storage volumes, salt caverns still support extremely large volumes of gas storage, and in a safe manner relative to high-pressure tanks or pipelines above the surface ([Figure 2](#)). In addition, the uncertainties associated with subsurface geological properties in depleted reservoirs or aquifers are of far lesser concern in engineered salt caverns which are essentially like cavities inside a homogenous salt layer.

Despite these benefits, feasible and cost-effective hydrogen storage in salt caverns are still challenging due to their restricted geographical locations (e.g., [Zivar et al., 2021](#)). Furthermore, due to the lower storage capability when compared with other porous media, more caverns have to be built to compensate for the large energy demand in some big cities. Therefore, if the storage capability of individual caverns can be further enhanced in a cost-efficient manner, UHS in salt caverns may significantly reduce the number of caverns needed while simultaneously reducing the operating costs over the life cycle of caverns.

The Concept of Enhanced Geological Hydrogen Storage

Hydrogen storage in salt caverns is carried out by compressing hydrogen gas or gas mixtures into the empty space within subsurface salt formations. The total amount of hydrogen that can be stored in and

later withdrawn from the cavern primarily depends on (1) volume of the empty space in salt cavern, (2) maximum and minimum operational pressures (and temperature) during the injection-withdrawal cycle, and (3) percentages of hydrogen (vs. cushion gas) in the gas mixtures stored in the cavern, in addition to other factors such as hydrogen loss. Here we present a novel method to increase the hydrogen storage capability of each cavern by filling the cavern with an economically-viable adsorbent material to increase the storage capability of individual caverns. Furthermore, the microporous infill material used here will provide extra pressure support to maintain cavern stability potentially over longer periods and minimize issues associated with leakage of brines/stored gas into surrounding formations. Importantly, the adsorbents used to fill the caverns could provide additional structural support to help maintain the integrity of the caverns and hence, to increase the total extractable hydrogen amount during each injection-withdrawal cycle (i.e., lower the minimal operational pressure). With this method, the number of caverns to be built for storing a given volume of hydrogen or maximizing the hydrogen storage for a given optimal salt cavern configuration can be reduced, optimizing the cost of salt cavern utilization and development.

Below, the following steps are proposed to enhance the hydrogen storage capability for salt caverns (Goteti and Patel, 2023).

Step 1: The void space in salt cavern can be generated by the process of solution mining during which a feed solvent is injected through a wellbore drilled into a subsurface salt formation (Figure 3, Warren, 2006). The brine generated subsequently by dissolving (leaching) the salt is then extracted to create the cavity within the salt formations. The process of feed solvent injection and brine extraction are repeated over multiple cycles until a cavern of desired dimensions is created in the subsurface. The generally preferred morphology for subsurface salt caverns is cylindrical, with several hundred meters high and 50 – 80 m in diameter. Fresh water or sea water may be used as feed solvent to create the cavity in the salt-rich formation.

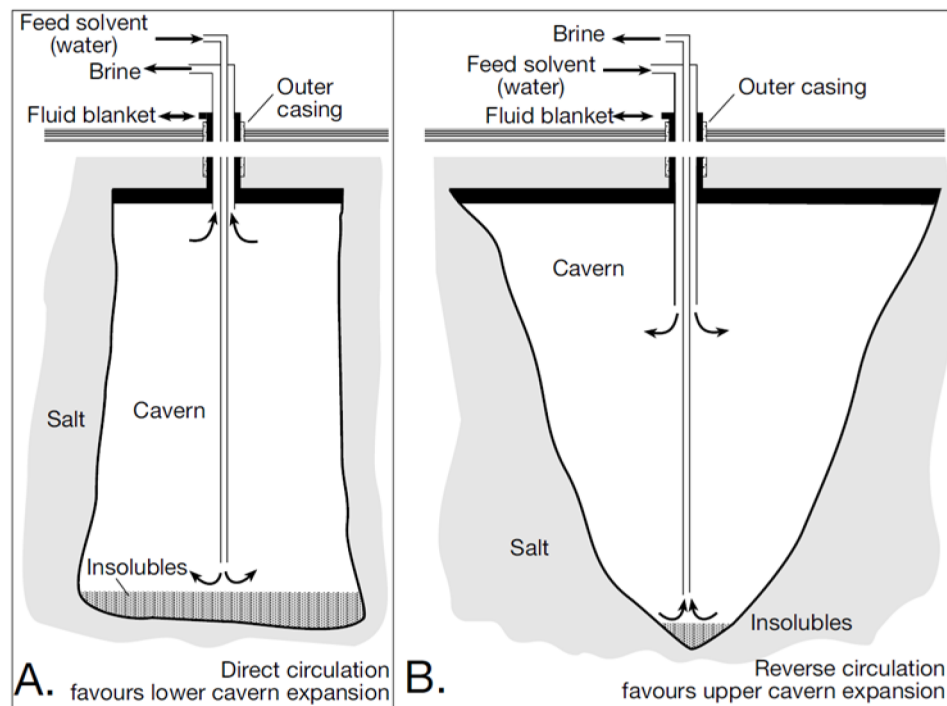


Figure 3—The solution mining process showing direct (a) and reverse (b) brine circulation for creating the cavity in a salt-rich formation (from Warren, 2006).

Step 2: Dehydration of the salt cavern will be carried out to further remove the remaining water or brine in the caverns. The compressed air can be used for removing contaminants from the salt caverns, as well as to protect the salt caverns from collapsing due to formation pressure (Figure 4).

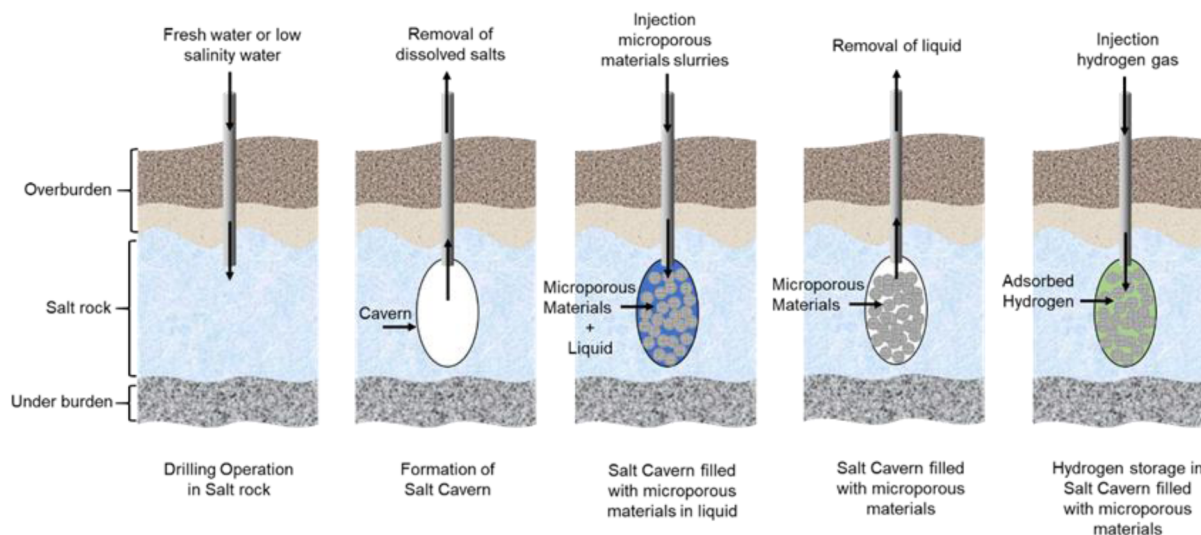


Figure 4—Process of hydrogen storage in salt cavern filled with micro-porous sorbent materials.

Step 3: A slurry of adsorbent materials—dispersed in either aqueous or non-aqueous liquids—is gradually pumped into salt caverns through multiple steps. These porous adsorbents may be organic, inorganic, or hybrid organic-inorganic materials, and can exist in amorphous, crystalline, polycrystalline, or mixed-phase forms. At each step of the process, the solid adsorbents are allowed to settle gravitationally. Excess liquid is then removed using compressed air. Once the salt caverns are fully filled with the adsorbent material, any remaining liquid can be evacuated using vacuum or pressure-driven methods. Alternatively, adsorbents may be introduced into the caverns in solid form—either as powders or pellets—via gas-phase transport. In both slurry and solid-state methods, excess liquid or gas can be expelled from the caverns using compressed air or other suitable gases.

Step 4: Once the salt caverns are filled with microporous adsorbent materials, hydrogen gas is injected under pressure from the surface. The hydrogen is compressed and stored within the porous structure of the adsorbents, in accordance with their theoretical storage capacity, which depends on the pressure and temperature conditions within the cavern. When the maximum hydrogen storage capacity is reached, the wellhead is sealed to enable long-term storage. Hydrogen can be released on demand by regulating surface pressure through controlled release valves. Desorption of hydrogen occurs as the pressure is reduced, allowing the gas to be extracted from the adsorbent material. This process enables efficient, high-capacity hydrogen storage in subsurface environments, leveraging the physical adsorption properties of the materials under controlled thermodynamic conditions.

Preliminary Results and Conclusion

To our best knowledge, there is no operational sites or public literature found on the hydrogen storage in an engineered cavern filled with adsorbent (i.e., subsurface storage in the presence of adsorbents), although various adsorbent materials have been proposed and tested as a promising hydrogen-storage materials for mobile applications (Schlapbach and Züttel, 2001). We have calculated theoretical capacity of hydrogen storage in cavern filled with different types of adsorbents (Table 1). The current result is outlining the next-generation hydrogen storage technologies that can reduce the number of caverns to be built for storing a given volume of hydrogen or maximizing the hydrogen storage for a given optimal salt cavern configuration. Because of enhanced H_2 storage in the presence of adsorbents, we can optimize the cost of salt cavern utilization and development.

Table 1—Performance of hydrogen storage in confined space filled with sorbents (from Goteti and Patel, 2023).

	Volumetric (kg H ₂ /L)		
	725 psi	1450 psi	2900 psi
Compressed H ₂ (empty tank)	0.004	0.0075	0.014
Compressed H ₂ (tank filled with Sorbent 1)	0.006	0.011	0.0175
Compressed H ₂ (tank filled with Sorbent 2)	0.005	0.009	0.0155
Compressed H ₂ (tank filled with Sorbent 3)	0.007	0.0112	0.0176*

Sorbent 1: carbon monolith (density: 1.07 g/cm³, surface area: 1050 m²/g, supplied by ATMI/Entegris inc.).

Sorbent 2: Activated carbon (density: 0.36 g/cm³, surface area: 3000 m²/g, Maxsorb3000 supplied by Kansai Coke & Chemical Co.). The hydrogen storage data at different pressures of Sorbent 1 and Sorbent 2 are obtained from the study by [Lozano-Castelló et al. \(2002\)](#).

Sorbent 3: Ni(dobdc) metal organic frameworks (density: 0.366 g/cm³, surface area 1029 m²/g) from [Kapelewski et al. \(2018\)](#).

* extrapolated H₂ capacity. These hydrogen storage capacities were measured on the surface in the tank or confined pressured space.

Figure 5 illustrates the difference in hydrogen storage capacity between an empty cavern and one filled with adsorbent materials, as a function of pressure. It is well established that gases are preferentially adsorbed onto solid surfaces, resulting in a higher concentration of gas molecules at the surface compared to the bulk phase. Microporous materials—such as activated carbons, mesoporous silicas, metal-organic frameworks (MOFs), microporous aluminosilicates (e.g., zeolites), and porous polymers—exhibit exceptionally high surface areas, typically ranging from 100 to 7000 m²/g. These materials demonstrate outstanding gas adsorption performance, particularly under high-pressure conditions. The enhanced hydrogen uptake is primarily attributed to the extensive microporous networks within these materials, which provide abundant adsorption sites and facilitate strong physisorption interactions with gas molecules.

We derived the theoretical hydrogen capacity of these materials when they are filled in the confined space at different pressures. If the salt cavern volume is about 1 million metric meters, we can store 4003 metric tons of hydrogen at 725 psi and 25 °C (Figure 5B). The hydrogen storage capacity reach to 30,027 metric tons at 1450 psi and 44,840 metric tons at 2900 psi at 25 °C. Upon loading the adsorbent (sorbent 1) in the empty space, the hydrogen storage capacity has increased within the same volume of the salt cavern (Figure 5C). The hydrogen storage capacity of salt cavern filled with sorbent 2 was observed to be 6005, 44,040, and 525,478 metric tons at 725, 1450, and 2900 psi, respectively at 25 °C. we have noticed 1.25-1.5 fold increase in the hydrogen storage capacity in the presence of sorbent 1. The calculated hydrogen storage capacity of another type of adsorbents, sorbent 2, is also derived to estimate the enhancement in hydrogen storage capacity when this adsorbent loaded in the salt cavern (Figure 5D). The salt cavern filled with sorbent 2 demonstrated hydrogen storage capacity of 7006 metric ton and 44,840 metric ton at 725 and 1450 psi at 25 °C. Hydrogen storage capacity can be enhanced to 1.5-1.75 fold compared to empty cavern of identical volume.

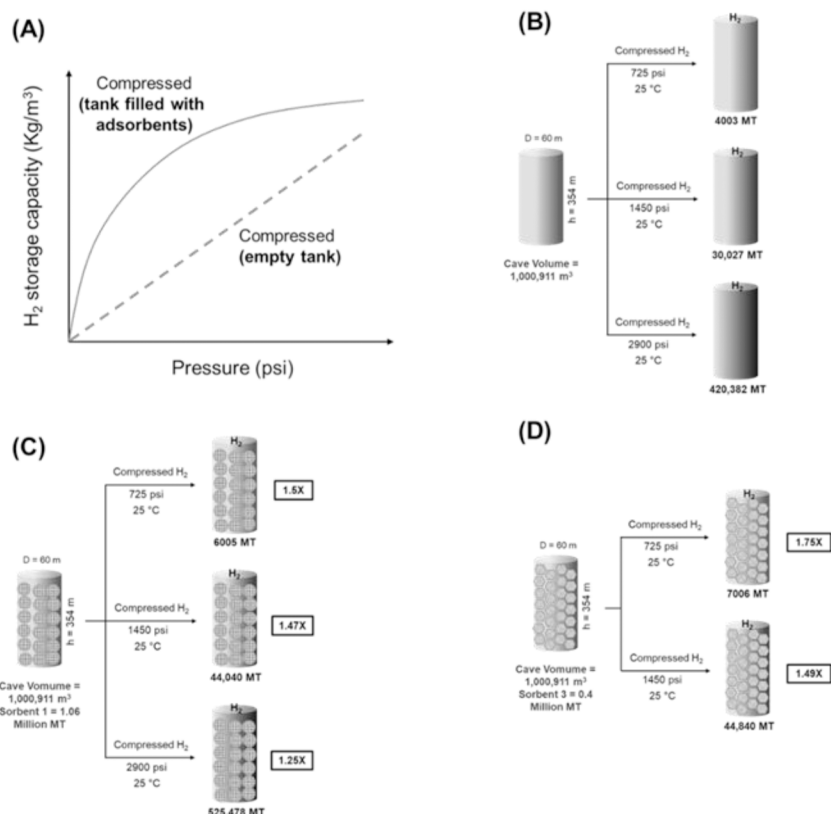


Figure 5—Approach to enhance the hydrogen storage capacity in Salt Caverns. (A) comparison on hydrogen storage capacity between empty space and space filled with adsorbents in confined condition. (B) calculated hydrogen storage capacity in empty space at three different pressures at $25\text{ }^\circ\text{C}$. (C) hydrogen storage capacity of empty space filled with Sorbent 1 at three different pressures at $25\text{ }^\circ\text{C}$. (D) hydrogen storage capacity of empty space filled with Sorbent 3 at three different pressures at $25\text{ }^\circ\text{C}$.

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